DATA ANALYTICS PROJECT

Churn Analysis and Customer Intelligence

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Project overview

Business challenge

In the competitive OTT market (Netflix, Hotstar, Prime), retention decides survival. I identified high-risk subscribers using customer demographics, subscription tiers and support escalations.

Tech stack

SQL and Python (numpy, pandas, sqlite3, matplotlib, seaborn) for data cleaning, feature engineering, analytics and behavioral visualization.

Milestones

Relational data extraction across tables, feature engineering (tenure, churn rate, customer age) and executive reporting of the findings.

Approach

1. **Connect** the SQL database to Python (pandas, sqlite3)
2. **Import** data with SQL queries across tables
3. **Clean** data types, column names, nulls and quality checks
4. **Engineer** calculated columns, transformations and filters
5. **Analyze** with aggregation, group by and pivot tables
6. **Visualize** with matplotlib and seaborn, then report insights

How churn is defined

I focused on three questions: **who** left or will leave, **why** they left based on earlier behavior, and **when** the danger zone begins.

Business type	Churn definition
SaaS	Subscription canceled
E-commerce	No purchase in 90 days
Adtech	No service or app use for 90 to 120 days
Streaming	Membership inactive
Telecom	Account terminated
Banking	No transactions for X months

Database: customer_churn

db_customer

- customerid
- name
- country
- state
- gender
- dob
- interests
- pincode

db_subscription

- customerid
- subscription_start_date
- subscription_type
- renewal_date
- plan_type
- contract_type
- cancellation_date
- cancellation_reason
- monthly_charges
- cltv
- churn_score

db_support

- customerid
- complaint_date
- escalations
- csat_score
- comment

Calculated metrics

KPI	Formula
Churn rate	Churned customers / total customers
Churn by plan type	Churn rate by plan_type (Basic, Standard, Premium)
Churn by state	Churn rate by country and state
Retention rate	1 - churn rate
ARPU	SUM(monthly_charges) / COUNT(active customers)
Average tenure	AVG(days from start date to cancellation date or today)
Revenue at risk	SUM(monthly_charges) where churn_score > 70
Escalation rate	SUM(escalations) / COUNT(complaints) x 100
Complaints per customer	COUNT(complaints) / COUNT(DISTINCT customerid)
Escalations vs churn	Churn rate with escalations ≥ 1 vs 0

Key insights

28.6%

Churn rate (retention 71.4%)

55.6% vs 8.3%

Monthly vs annual contract churn

2,047

CLTV lost to churn

- Most churn comes from the Basic plan, so the revenue impact is limited.
- Churn peaked in Sep 2024, and Karnataka is the most affected state.
- Average tenure is 1,451 days and ARPU is Rs 18.8.
- Total revenue is 395; revenue lost to churn is 74 (18%).

Recommended actions

Investigate Karnataka
Check for price increases, user complaints or technical issues.

Review Basic plan changes
Find out whether pricing or features changed, especially in September.

Watch competitors
At least one user moved to a competitor, so benchmark offers.

Prioritize high-risk users
Rank High and Medium risk customers by CLTV, review complaints, then reach out by email, SMS or call.

Project impact

Churn and revenue: Built an end-to-end churn analytics pipeline for an OTT subscription dataset (20+ KPIs). Found a 28.6% overall churn rate, with monthly-contract subscribers churning at 55.6%, 6.7 times the 8.3% annual rate. Tied $73.94/month of MRR leakage and $2,047 of CLTV erosion to six at-risk customers, supporting a contract-migration retention strategy.

Risk scoring: Segmented customers into risk tiers using tenure, plan type and support escalation signals, and recommended prioritizing Premium annual retention over Basic monthly acquisition.

Support intelligence: Joined complaint, escalation and CSAT data with subscriptions to link escalations to churn and break down cancellation drivers.